Predicting Mental Health Risk Among Tech Employees

A Hybrid Machine Learning Approach

Mental Health Challenges in the Tech Industry

- Mental health issues such as anxiety, depression, and burnout are rapidly rising among technology professionals.
- Many employees actively avoid seeking help due to persistent workplace stigma.
- Organizations currently lack predictive tools to detect early mental health risks before they escalate.
- Poor mental health directly leads to reduced productivity and increased employee turnover.

Key Insight: Early detection of mental health risks can significantly improve employee wellbeing and organizational productivity.

Project Objectives: From Data to Support

Analyze: Evaluate employee mental health survey data from the tech industry.

Identify: Spot critical patterns in workplace environment and mental health behavior.

Cluster: Group employees based on shared mental health risk indicators.

Predict: Build a predictive model to detect employees likely to seek treatment.

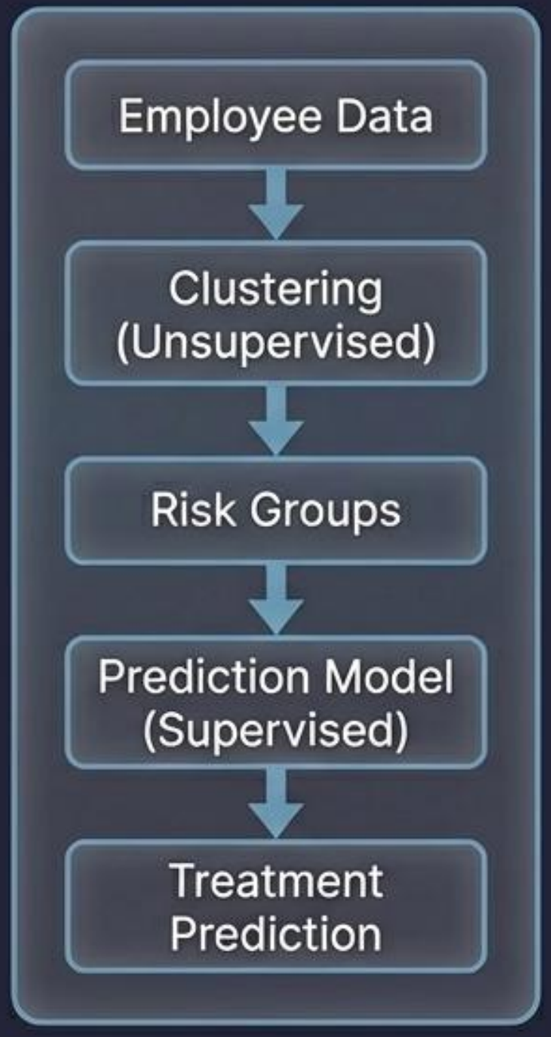
Support: Empower organizations to improve workplace mental health programs.

Data → AI Model → Prediction → Workplace Support

A Hybrid Machine Learning Approach

Unsupervised Learning

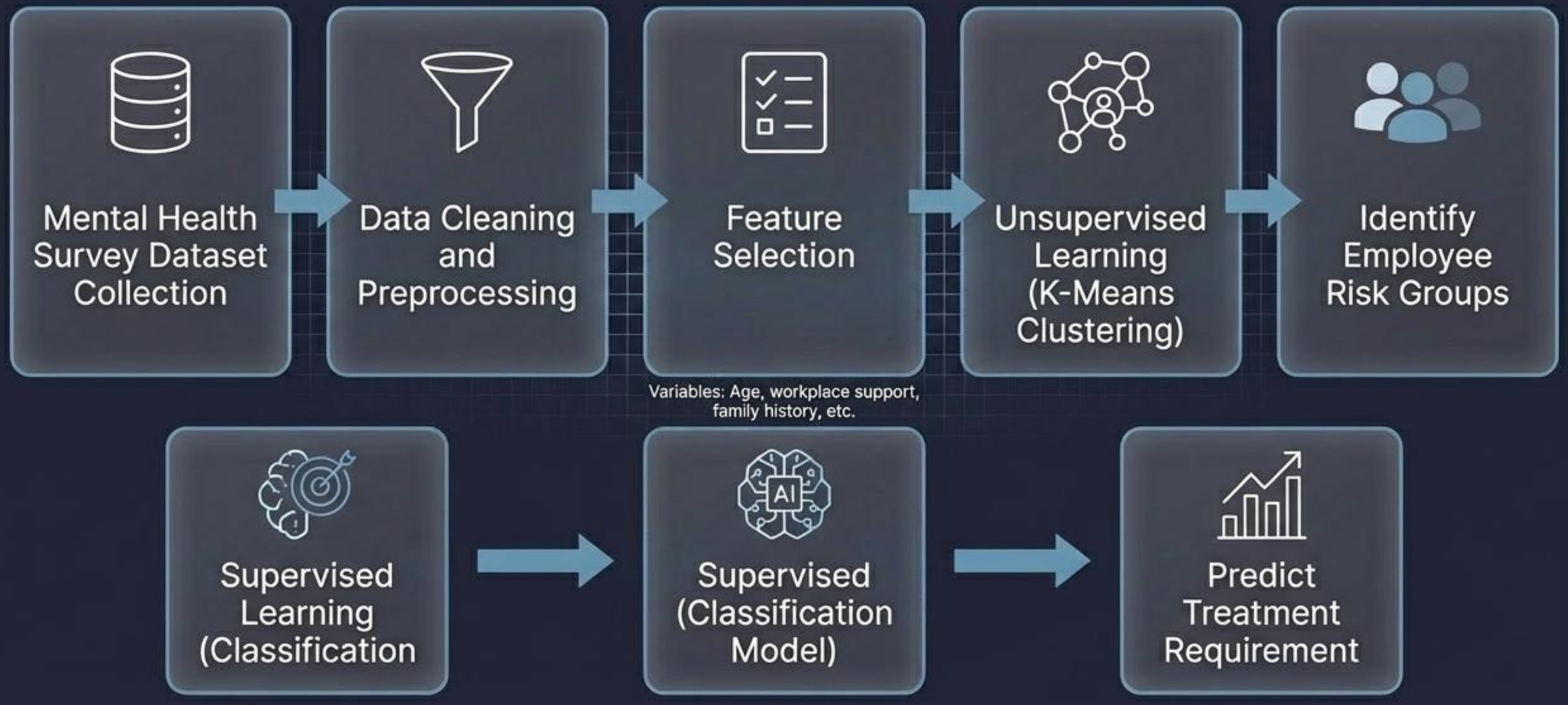
- Uses clustering techniques such as K-Means
- Groups employees based on mental health patterns
- Identifies hidden behavioral groups
- Example clusters: High stress employees, Moderate stress employees, Low risk employees



Supervised Learning

- Predicts whether employees will seek mental health treatment
- Algorithms used: Random Forest, Logistic Regression, Decision Tree

Hybrid Model Workflow



Impact and Real-World Benefits

Employee Wellbeing

- Early identification of mental health risks
- Improved workplace support systems
- Reduced stigma around mental health

Organizational Benefits

- Higher overall productivity
- Reduced employee burnout
- Lower employee turnover

Social Impact

- Better mental health awareness
- Healthier, more empathetic workplace culture
- Data-driven HR decision making

AI-driven mental health prediction systems can help organizations create healthier and more supportive workplaces.